TABLE III
WORKING CAPITAL PER MONTH

1. STAFF AND LABOUR

Particulars	Quantity	Salary (₹)	Total salary (₹)
Manager	1	20,000	20,000
Sales engineer	1	17,000	17,000
Accountant cum clerk/typist	1	15,000	15,000
Peon	1	5,000	5,000
Watchman	1	5,000	5,000
Skilled worker	5	8,000	40,000
Unskilled worker	6	4,500	27,000
Total salary per month			129,000
Add perquisites @ 15% of salary			19,350
TOTAL			**148,350**

2. RAW MATERIALS REQUIREMENT

Particulars	Indian/Imported	Cost for producing 80,000 resistors per month (₹)
Viterous enamel/silicon-based resin (80kgs)	Imported	56,000
Strips (80kgs)	Imported	80,000
Resistance wires (40kgs)	Imported	75,000
Caps and beads	Imported	20,000
Consumables like hardener, rubber sheet and packaging	Indian LS	10,000
TOTAL		**241,000**

3. UTILITIES

Particulars	Average cost (₹)
Power	4,000
Water	1000
Total	**5,000**

4. OTHER CONTINGENT EXPENSES

Item	Amount (₹)
Rent	60,000
Postage and stationery	3,000
Repair and maintenance	3,000
Telephone/telex/fax charges	2,000
Transport and conveyance charges	6,000
Advertisement and publicity	5,000
Insurance	1,000
Miscellaneous expenditure	2,000
TOTAL	**82,000**
TOTAL WORKING CAPITAL PER MONTH (1+2+3+4)	**476,350**

TABLE IV
TOTAL CAPITAL INVESTMENT

Total capital investment	Cost (₹)
Fixed capital	459,900
Working capital for three months (476,350×3)	1,429,050
Total	**1,888,950**

TABLE V
COST OF PRODUCTION PER ANNUM

Financial analysis	Cost (₹)
Total recurring expenditure (476,350×12)	5,716,200
Depreciation on machinery/equipment @ 10%	32,900
Depreciation on office equipment/furniture @ 20%	12,000
Depreciation on jigs/fixtures @ 25%	4,500
Interest on total capital investment @ 16%	73,584
Total	**5,839,184**

TABLE VI
TURNOVER PER ANNUM

Item	Quantity	Rate per unit (₹)	Total sales (₹)
Wire-wound resistors (different values)	960,000	7.5	7,200,000

To create a high resistance, wire diameter needs to be small and the length long. Therefore wirewound resistors are mainly produced for lower resistance values.

For low power ratings, a very thin wire is used. Handling of the wire is, for this matter, critical. Any damage may sever contact. After winding, the wire is well protected from access to moisture to prevent electrolytic corrosion.

Next to precision, there are also wirewound resistors with high power rating for 50W or more. These have a different construction. Compared to other resistor types such as metal film, wire diameter is relatively big and therefore more robust.

Raw materials and equipment

A wirewound resistor is an electrical passive component that limits current. The resistive element exists out of an insulated metallic wire, which is wound around a core of non-conductive material. The wire material has high resistivity, and is usually made of an alloy such as nickel-chromium (nichrome) or a copper-nickel-manganese alloy called manganin. Common core materials include ceramic, plastic and glass.

Wirewound resistors are the oldest type of resistors that are still manufactured today. These can be produced very accurately, and have excellent properties for low resistance values and high power ratings.

These are mainly produced with alloys, since pure metals have a high temperature coefficient of resistance. However, for high temperatures, pure metals such as tungsten are used.

Temperature coefficient is a sign of how much the resistance will change as temperature changes. TCR (temperature coefficient of resistance) is measured in units of ppm/°C. If a manufac-